

Interactive Visual Analytics for Generating and Exploring LLM-Predicted Protein-Protein Interactions

Allison Austin
amaustin@ucdavis.edu
University of California, Davis
California, USA

Shilpika
amaustin@anl.gov
Argonne National Laboratory
Illinois, USA

Aditya Tanikanti
aditya.tanikanti@gmail.com
Argonne National Laboratory
SLAC National Accelerator
Laboratory
USA

Venkatram Vishwanath
venkat@anl.gov
Argonne National Laboratory
Illinois, USA

Michael E. Papka
papka@anl.gov
Argonne National Laboratory
University of Illinois Chicago
Illinois, USA

Kwan-Liu Ma
klma@ucdavis.edu
University of California, Davis
California, USA

Abstract

Protein-protein interactions (PPIs) are fundamental to understanding cellular function and disease mechanisms. While large language models (LLMs) show significant promise in predicting novel interacting proteins, large-scale evaluation of these predictions remains a challenge. Validation typically requires cross-referencing predictions against molecular databases, which often omit biologically plausible but formally untested interactions. Current workflows for exploring these "hidden" or predicted interactions are fragmented, requiring researchers to combine text-based LLM outputs with separate structural viewers, sequence alignment tools, and network visualizations. In this paper, we present a visual analytics system designed to support real-time generation, exploration, and evaluation of LLM-generated PPIs within a single, integrated environment. Our system enables users to dynamically query LLMs and visualize the resulting predictions in real-time. We present two use cases centered on the PD-1 axis, identifying novel predictions across multiple evidence channels and successfully distinguishing between functionally-associated protein clusters from direct physical interactors. By unifying rapid LLM inference with rigorous biological scoring and high-performance visualization, our system provides an end-to-end framework for interpreting large-scale, AI-generated biological networks.

CCS Concepts

• **Human-centered computing** → **Visualization systems and tools.**

Keywords

protein-protein interactions (PPI), large language models, PPI prediction, visual analytics, network visualization, explainable AI

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1 Introduction

Protein-protein interactions (PPIs) are essential to understanding cellular function disorders and diseases. The study of PPI networks has a broad range of applications, including drug discovery, disease diagnosis, and drug design [33]. However, PPI data is being rapidly accumulated due to advancements in PPI network detection. For instance, the STRING database currently contains information on 59 million proteins across over 12,000 organisms [29]. Given the increasing wealth of PPI data, large-language models (LLMs) have been employed for various tasks such as protein function prediction [31], protein engineering [11], and interaction prediction [23].

Although the use of LLMs to generate protein interactions has been extensively studied in recent years [23], practical evaluation of interaction predictions remains a challenging task. While PPI networks generated by LLMs contain many true positive interactions, these predictions require further assessment. Current methods are limited to comparing predictions to empirically-derived results or assessing genetic, sequential, and/or structural similarity between the predicted interacting proteins [22]. However, this approach is limited by the inherent bias and incompleteness of primary repositories, as well as the data discrepancies that exist across several PPI databases [6]. Additionally, practical tools that support user understanding of the feasibility and origin of LLM-based predictions are limited. Existing web-based tools for visualizing PPI networks, such as Cytoscape [26], primarily function as static rendering engines for curated datasets. They lack support for real-time LLM integration and, crucially, do not provide the granular evidence required to distinguish between direct physical docking and indirect functional convergence.

To account for these gaps, we explore the real-time evaluation of PPIs generated by LLMs using interactive visual analytics (VA). We propose a quantitative framework that allows bioinformatics



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experts to gain insight on the characteristics of both established and novel PPI predictions. Our system incorporates a computational pipeline that produces a comprehensive score for each PPI prediction that integrates structural evaluation with functional and topological measures. Our implementation is designed to scale efficiently, allowing for PPI candidates to be generated and evaluated in near real-time. Our main contributions are: (1) A technical pipeline for generating, parsing, and evaluating PPI predictions, and (2) A scalable, interactive interface¹ that supports experts in evaluating predicted interactions.

2 Related Work

2.1 Protein-Protein Interaction Prediction

Several recent studies have evaluated the efficacy of various LLMs for predicting protein behavior, with a focus on interaction prediction [10, 20]. These works commonly leverage established biological knowledge bases (e.g., STRING [29], KEGG [15], UniProt [2]) to benchmark LLM predictions against known binding proteins. However, an interaction's absence from these knowledge bases is not conclusive proof of a false prediction, as it may reflect limited experimental data. Additionally, interactions present in these knowledge bases can suffer from inflated confidence scores. For instance, STRING's combined score represents functional associations rather than implying a physical interaction [29], therefore, proteins that are frequently co-mentioned in literature from shared pathway involvement may receive high scores despite a lack of physical interaction. Therefore, assessing PPIs in large-scale networks requires a multi-channel validation approach. To enable a more nuanced assessment, prior work has employed computational metrics synthesizing Gene Ontology (GO) information (Biological Processes (BPs), Molecular Functions (MFs), and Cellular Components (CCs)) [5] alongside structural and sequence similarity measures [14]. Since proteins involved in the same cellular processes are more likely to interact [32], measuring the semantic similarity between BP annotations—particularly through the Resnik method [21, 33]—has emerged as a highly effective metric. Beyond semantic and text-mining evidence, a significant body of research focuses on predicting the physical docking of proteins using amino acid sequences [18]. One example is D-SCRIPT [28], which utilizes a pre-trained protein language model to generate contact-map-based interaction probabilities.

Efforts to unify these disparate evidence channels have led to the development of pipelines such as PEPPi [8], which aggregates structural similarity, functional association, and sequence data into a single interaction probability. While PEPPi demonstrates the power of multi-modal evidence for high-accuracy screening, it remains a "black-box" predictive framework. Specifically, it fails to provide a mechanism for deconvolving the underlying evidence types or assess the biophysical or functional drivers of a prediction.

2.2 Visualizing Protein-Protein Interactions

With the emergence of deep-learning models for PPI prediction, there is a growing need to incorporate these models into bioinformatics workflows at larger scales. Graph-based visualizations are widely used to analyze PPI networks [24], and a broad range

of software tools for visually exploring biological networks have been surveyed [1]. Existing tools can be broadly grouped into two categories: (1) general-purpose network visualization tools, and (2) network visualization tools specifically designed for biological data analysis.

General-purpose graph visualization tools like Gephi [7], and Graphistry [13] are capable of rendering large-scale PPI networks and performing complex topology analysis on these graphs, but are not optimized for analyzing biological data. Popular domain specific tools for molecular interaction networks such as Cytoscape [26] provide rich functionality for integrating heterogeneous biological data and exploring large, complex networks. While these tools provide a robust ecosystem for biological network analysis, they are primarily designed for the visualization of validated, curated datasets. Cytoscape's architecture is built around the manipulation of static PPI networks, where edges represent established literature links or experimental results. Specialized tools like HIPPIE [3] provide high-confidence, context-specific PPI prediction networks with evidence derived from functional associations and database evidence. However, HIPPIE does not leverage advanced deep-learning models to evaluate novel predictions, nor does it consider structural compatibility between proteins. Similarly, computational pipelines like the aforementioned PEPPi [8] provide standalone web servers for querying and visualizing predicted networks, but offer limited explainability in their underlying architectures. Our work extends the capabilities of these systems by targeting the evaluation of predicted protein behavior at scale.

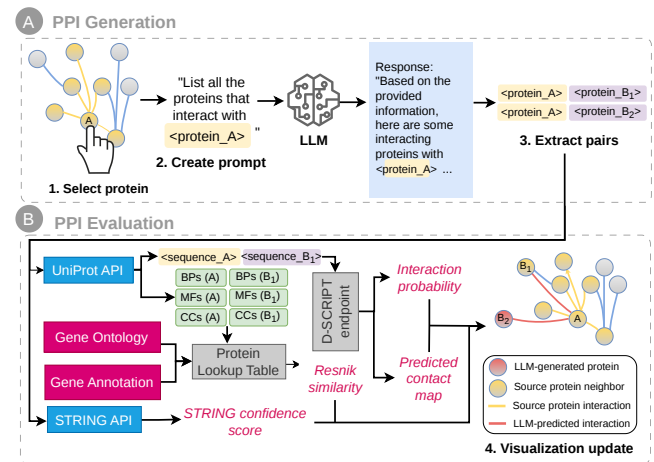


Figure 1: Pipeline for generating and analyzing protein-protein interaction predictions.

3 System Overview

3.1 Prediction Pipeline

We employ a dynamic, user-driven discovery workflow for PPI generation. When a user selects a *seed* protein, the framework initiates a three-stage analysis: 1) LLM-based candidate generation, 2) identifier mapping and validation, and 3) structural and functional scoring. Details regarding our system implementation, including the exact model and prompts used in our pipeline, can be found in Appendix C.

¹<https://github.com/allisonaustin/protein-graph-vis>

3.2 Prediction Scoring and Annotation

Once a predicted pair (p_1, p_2) is validated as a real biological entity, we assign prediction scores using three complementary modalities: (1) **Database Evidence**, querying the STRING API [29] for each candidate protein pair (p_1, p_2) and attaching the corresponding STRING confidence scores when available, (2) **Structural Compatibility**, evaluating the likelihood of a physical binding between two proteins using the D-SCRIPT framework [28], and (3) **Functional Similarity**, calculating the semantic similarity over GO annotations [5]. Together, these scores (STRING, D-SCRIPT, Resnik) provide interpretable profiling for every predicted interaction. Table 1 provides a guide to interpreting biologically relevant score patterns generated by our system.

Table 1: Signal Interpretation. Significance of high $\uparrow$ & low $\downarrow$ scores for STRING (S), D-SCRIPT (D), and Resnik (R).

Pattern	Significance
$S \downarrow D \downarrow R \downarrow$	Functional association or physical interaction is unlikely.
$S \downarrow D \uparrow R \downarrow$	Structurally similar, unlikely to be functionally related.
$S \uparrow D \uparrow R \downarrow$	Verified physical interaction, low functional similarity.
$S \downarrow D \downarrow R \uparrow$	Functional association, low likelihood of physical contact.
$S \uparrow D \downarrow R \uparrow$	Functional synergy (e.g., co-therapy targets).
$S \downarrow D \uparrow R \uparrow$	Strong candidate for experimental validation.
$S \uparrow D \uparrow R \uparrow$	Useful for benchmarking/identifying well-studied pathways.

3.3 Interface

The interface (Fig. 2-2) facilitates a "search-predict-evaluate" workflow across three coordinated modules: a 3D Network View (Fig. 2-2), Information Panel (Fig. 2-3), and Summary Table (Fig. 2-1).

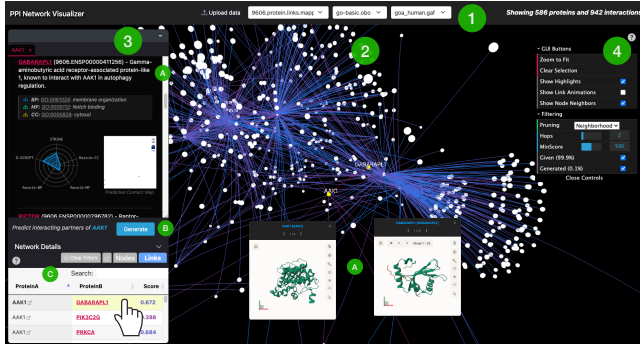


Figure 2: The UI of our visual analytics tool, consisting of an information panel (1), an interactive 3D network of proteins and interactions with controls (3). 1A) contains the LLM interaction component, where users can generate and assess PPI predictions of a selected protein in the network. 1B) contains all of the PPI predictions and score details.

3.3.1 3D Network View. The primary view provides a 3D force-directed layout representing proteins (nodes) and their interactions (edges) (Figure 2-2). While 2D networks are effective for analyzing small PPI networks, they suffer from significant visual 'clutter' and edge crossing when representing dense, high-degree clusters. By

using a 3D network layout, our system increases available spatial resolution, allowing for a more effective separation of visual clusters. Users can interact with the network using zoom, pan, and rotation controls for spatial navigation. Hovering over nodes will reveal protein identifiers and interaction neighborhoods, while clicking nodes will highlight their direct interaction partners. When the user selects a *seed* protein, we use categorical color coding to distinguish interaction sources: yellow for edges validated by database evidence or user-provided uploads, and deep pink for LLM-predicted interactions. For the remaining graph, interaction confidence is mapped to a continuous color scale from pink (low) to blue (high). Users can navigate via 3D rotation and zoom, or isolate neighborhoods to n -hops to focus on specific clusters. For proteins with known structures, users can right-click on a node to open the corresponding 3D molecular structures rendered using Mol* Viewer [25], enabling direct visual inspection of structural similarity between candidate partners (Figure 2-2A).

3.3.2 Information Panel. The information panel (Figure 2-3) complements the 3D network by providing quantitative summaries and detailed metadata views of the PPIs. Users can generate PPI candidates by selecting a "seed" protein directly from the network or table and the LLM response is parsed, evaluated, and visualized in the prediction assessment module (Figure 2-3A). The validated proteins generated by the LLM are added to the network in real-time, and these proteins/interactions are labeled and colored as "generated" in the network, to distinguish from user-added proteins or proteins provided in the base PPI network. When the user generates interactions via the LLM module, the panel dynamically renders evidence profiles to provide structural and functional grounding for each prediction (Figure 4).

To validate these textual claims, the system calculates and displays the Most Informative Common Ancestors (MICAs) of the GO terms (BP, MF, CC) shared by the protein pair (Figure 4-B). The information content (IC) of each of these annotations is visualized in a radar plot (Figure 4-C) along with the D-SCRIPT interaction probability score and the STRING database confidence score (shown as 0 if unavailable). To assess structural compatibility, the window includes the predicted inter-protein residue-contact map (Figure 4-D). This heatmap represents the probability of interaction between all residue pairs between the candidate protein pair, with higher probabilities shown in dark blue.

3.3.3 Summary Table. The final component of this panel is a comprehensive list of all proteins, link counts, their interaction partners, and confidence scores (Figure 2-1C). Scores for predicted interactions are computed by taking the maximum of the collected scores. Users can search and sort by specific protein names, GO terms, or score thresholds to prioritize candidates for interaction prediction or experimental follow-up. Hub proteins in the network are easily identifiable by sorting proteins by their link counts.

4 Case Studies

To evaluate the practical utility of our system, we integrated the human PPI dataset from the STRING interaction network data [29] and mapped ENSP identifiers to HGNC symbols [30] for improved user familiarity. We render a maximum 5,926 proteins and 10,006

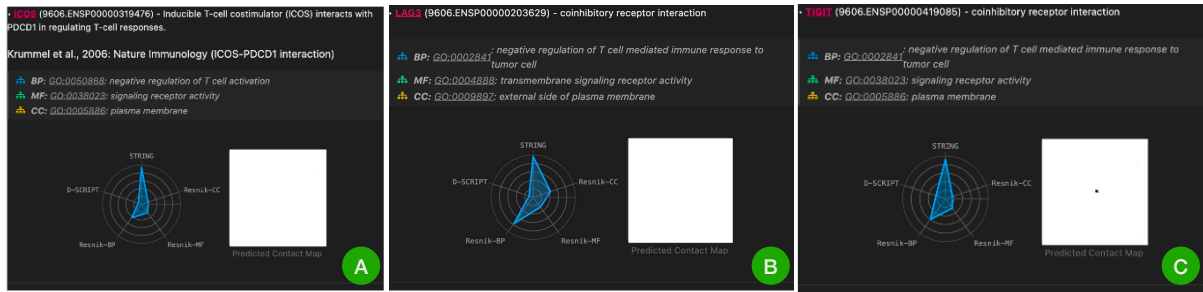


Figure 3: Visual summaries of predicted interacting partners of PDCD1 (PD-1) protein.

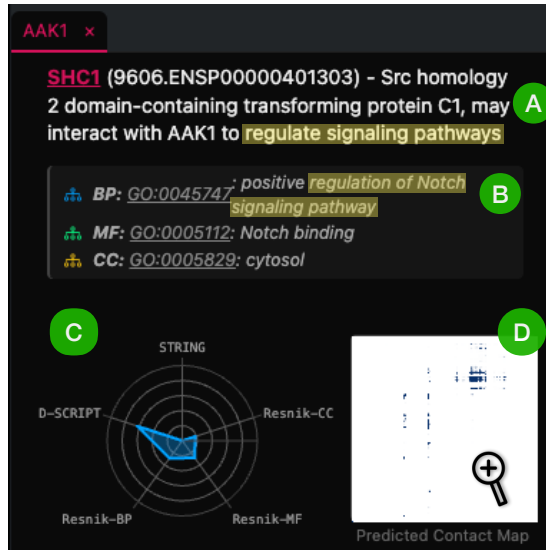


Figure 4: Evaluating AAK1-SHC1. The panel displays: (A) LLM-derived reasoning; (B) shared GO terms; (C) radar plot of interaction scores; and (D) a residue-contact map.

interactions in our base network to reduce initial load time for our experiments. This base network was dynamically extended via LLM-driven discovery. We highlight the usability of our system in two key use cases: (1) identifying distinct functional roles among well-studied, highly confident PPIs, and (2) identifying potential candidates among under-studied proteins.

4.1 Use Case 1: Mechanistic Dissection of the PD-1 Predictions

One of the key functionalities of our system is the ability to visually distinguish direct physical docking partners and indirect functional signaling partners. For instance, we queried the predicted interactions for the PDCD1 protein, and repeatedly got TIGIT, ICOS, and LAG-3 as interacting partners. When examining these partners on the surface, their combined STRING scores are high (0.846, 0.998, and 0.965, respectively), yet a review of the supporting literature provided by STRING indicates this score is driven by functional co-expression and clinical correlation rather than physical docking. These genes are frequently analyzed as a unified "checkpoint set" due to their shared role in T-cell exhaustion [17], leading to

high statistical correlations in patient data [19] that do not imply physical binding. Both proteins inhibit T-cell activation by disrupting CD226 signaling, but through distinct mechanisms [17]. The low D-SCRIPT score (Figure 3) confirms this low probability of a direct physical docking. This is a critical distinction that suggests that these targets should be approached as a combination therapy (targeting two different points) rather than a bispecific antibody (targeting a single complex).

We see a similar pattern in the PDCD1-ICOS and PDCD1-LAG3 predictions (Figure 3). In the case of ICOS, the near-perfect STRING score (0.988) reflects its status as the defining co-expression marker for T follicular helper (Tfh) cells rather than a physical interface. Similarly, the LAG-3 score is heavily weighted by 'text-mining' and 'co-expression' channels due to the clinical prominence of PD-1/LAG-3 combination therapies. While these receptors are frequently co-expressed on tumor-infiltrating lymphocytes, a review of the underlying evidence confirms that their 'interaction' is characterized by a functional cooperation in dampening T-cell responses rather than a physical complex formation, as their primary, high-affinity functional ligands are distinct (MHC II for LAG-3 and PD-L1 for PD-1) [4]. The discrepancy between the high STRING scores and the low D-SCRIPT scores (Figure 3) highlights a bias in association databases toward functional synergy over structural docking.

4.2 Use Case 2: PPI Discovery Using Evidence Synthesis of PD-1 Predictions

We demonstrate the system's ability to evaluate novel PPI predictions that extend beyond curated databases. Focusing on PDCD1, the LLM predicted LILRB4 (also known as ILT3) as a potential interactor—an association currently absent from the base STRING interaction network. This interaction exhibits a high Resnik-BP score (0.71), reflecting the pair's shared role in immune checkpoint inhibition and negative regulation of tolerance induction (Figure 5-A). The topological assessment of these proteins in the network context (Figure 5-B) suggests that LILRB4 serves as a functional bridge between two network hubs. Notably, while LILRB4 is missing from the original neighborhood of PDCD1, it maintains a direct connection to CD4, which is directly linked to PDCD1. This highlights the ability of our system to identify hidden functional links between proteins in large networks. We confirmed this link between myeloid-driven suppression and the T-cell centric axis in one study, which found that LILRB4 is highly expressed on monocytic myeloid-derived suppressor cells, where it orchestrates immune

evasion by suppressing T-cell activation in the tumor microenvironment [12].

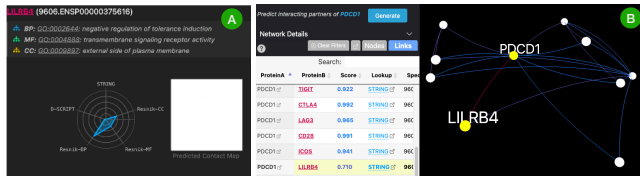


Figure 5: Depicting novel interaction (PDCD1–LILRB4) summary (A) and relevance in relation to neighboring proteins (B). The LLM provided no reasoning for this interaction, but share a highly specific biological process (*GO:0002644*). They are most likely *functionally* associated rather than direct binding partners given the low D-SCRIPT score (0.04).

While the probability of a direct physical docking between PDCD1 and LILRB4 is low (D-SCRIPT=0.04), their high ontogenic similarity and shared connectivity to the CD4 hub indicate they are highly correlated and co-expressed in immunosuppressive environments. By surfacing LILRB4 as a parallel therapeutic target, the system demonstrates its utility in identifying "bypass mechanisms" that may contribute to anti-PD-1 resistance, moving beyond simple database lookup toward mechanistic discovery.

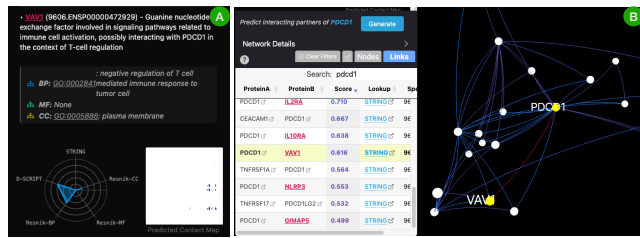


Figure 6: Depicting novel interaction (PDCD1–VAV1) summary (A) and relevance in relation to neighboring proteins (B). While not in the STRING database, this protein pair shares a specific biological process (*GO:0002841*) and have structural compatibility (D-SCRIPT=0.615).

Beyond functional bridging, the system also identifies novel candidates that occupy the same proximal signaling neighborhood. The LLM predicted a interaction between PDCD1 and VAV1, a guanine nucleotide exchange factor central to T-cell antigen receptor (TCR) signaling [16]. While this pair is also absent from the base STRING human interaction network, our multi-dimensional assessment reveals a specific shared biological process (*negative regulation of T cell mediated immune response to tumor cell*), and moderate structural compatibility (Figure 6-A). Topologically, VAV1 is positioned within the immediate neighborhood of the T-cell receptor complex (Figure 6-B). The predicted contact map for this pair (Figure 6-A) exhibits high probability clusters between the C-terminal signaling tail of PDCD1 and the N-terminal regulatory domains of VAV1. This structural affinity could reflect their shared spatial recruitment to TCR microclusters, where PDCD1 signaling has been shown

to specifically attenuate VAV1 phosphorylation [34]. By surfacing such candidates, the system provides a mechanistic map of the "inhibitory neighborhood," identifying proteins that are structurally positioned to be affected by the PD-1 signaling axis during T-cell inactivation.

Overall, our use cases demonstrate the utility of our system in integrating structural prediction, functional ontology, and LLM-driven literature synthesis. By deconvolving high-confidence database scores into their physical and functional parts, we found that many high-scoring interactions in the PD-1 axis—such as LAG3 and TIGIT—represent proximal signaling collaborators rather than direct docking partners. Furthermore, the identification of the PDCD1–LILRB4 and PDCD1–VAV1 links demonstrates that the system can proactively suggest novel therapeutic targets that are topologically valid but currently absent from curated repositories.

5 Discussion

The results from our case studies underscore a critical gap in current bioinformatics workflows: the conflation of "association" with "interaction." While the STRING database serves as a premier repository for protein relationships, its 'combined' score can be misinterpreted as a metric of physical binding. As demonstrated in the PDCD1–TIGIT and PDCD1–LAG3 axes, a high STRING score (0.85) can be driven almost entirely by text-mining and co-expression data. Our system provides the necessary mechanistic deconvolution to reveal that these proteins, while highly functionally associated, lack the structural compatibility required for direct docking.

A natural comparison to our tool is Cytoscape [26], the industry standard for PPI network visualization. Cytoscape primarily functions as a static rendering engine for pre-existing datasets. While it can import STRING networks, it lacks the dynamic synthesis and real-time inference capabilities of our platform. Specifically, Cytoscape does not generate novel hypotheses, perform on-the-fly structural validation, or synthesize multi-modal evidence to distinguish physical and functional interactions. By moving beyond a simple graph lookup, our tool acts as an active collaborator in the discovery process, filtering out hallucinations caused by text-mining bias and highlighting high-probability structural targets. Our framework is also highly configurable; users can integrate alternative databases, ontology versions, scoring approaches, and LLM backends. This enables adaptation to evolving models, data sources, and evaluation needs in computational biology.

While LLMs remain valuable for specific tasks such as information retrieval and summarization, their utility is often limited by the difficulty in extracting structured biological data from unstructured natural language. Current LLM-based PPI research highlights that sophisticated prompt engineering and iterative refinement are required to ensure that model outputs adhere to strict scientific ontologies [9]. Our system can address this by acting as a structural "filter" for these prompts. Additionally, LLM reasonings lack explicit indicators of protein research coverage or "confidence" metrics. This is particularly problematic given that well-studied proteins benefit disproportionately from available training data [20]. This results in varying network topologies, which are inherently sensitive to the underlying LLM's architecture and training biases [20]. Our framework functions as a comparative tool for using grounded evidence

to compare model outputs, identifying which models favor specific pathways or exhibit higher hallucination rates for certain proteins.

6 Limitations and Future Work

While the current system significantly enhances PPI discovery, there remain several improvements. A primary technical restraint is the sequence length limitation of the D-SCRIPT model. Since D-SCRIPT relies on a pairwise residue-comparison matrix, its computational complexity scales quadratically ($O(n^2)$) with protein sequence length. To address this, future iterations of the platform will explore the integration of other frameworks designed for linear-time prediction of physical protein interfaces.

The network topology is inherently sensitive to the underlying LLM's training biases. Our framework functions as a comparative tool for using grounded evidence to compare model outputs, identifying which models favor specific pathways or exhibit higher hallucination rates for certain proteins. Future work is needed to streamline this comparison by extending the visualization to compare model outputs in one view.

Local tests confirmed no performance degradation for networks up to ~6,000 nodes and ~10,000 edges, though further optimization is needed to ensure seamless performance at larger scales. The "cross-database discrepancy" problem—where protein databases report non-overlapping interaction sets, remains a significant hurdle for researchers [6]. We plan to implement a multi-source consensus engine that can highlight these discrepancies in real-time.

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References

- [1] Giuseppe Agapito, Pietro Hiram Guzzi, and Mario Cannataro. 2013. Visualization of protein interaction networks: problems and solutions. *BMC Bioinformatics* 14, 1 (2013), S1. doi:10.1186/1471-2105-14-S1-S1
- [2] S. Ahmad et al. 2025. The UniProt website API: facilitating programmatic access to protein knowledge. *Nucleic Acids Research* 53 (2025), W547–W553. doi:10.1093/nar/gkaf394
- [3] G. Alanis-Lobato, M. A. Andrade-Navarro, and M. H. Schaefer. 2016. HIPPIE v2.0: enhancing reliability of PPI networks. *Nucleic Acids Research* 45, D1 (2016), D408–D414. doi:10.1093/nar/gkw985
- [4] Ana C. Anderson, Nicole Joller, and Vijay K. Kuchroo. 2016. Lag-3, Tim-3, and TIGIT: Co-inhibitory Receptors with Specialized Functions in Immune Regulation. 44, 5 (2016), 989–1004. doi:10.1016/j.immuni.2016.05.001
- [5] M. Ashburner et al. 2000. Gene Ontology: tool for the unification of biology. *Nature Genetics* 25 (2000), 25–29. doi:10.1038/75556
- [6] Akhilesh Kumar Bajpai et al. 2020. Systematic comparison of the protein-protein interaction databases from a user's perspective. 103 (2020), 103380. doi:10.1016/j.jbi.2020.103380
- [7] Mathieu Bastian, Sebastien Heymann, and Mathieu Jacomy. 2009. Gephi: An Open Source Software for Exploring and Manipulating Networks. In *Proceedings of the International AAAI Conference on Web and Social Media (ICWSM)*. 361–362. <https://gephi.org>
- [8] Eric W. Bell, Jacob H. Schwartz, Peter L. Freddolino, and Yang Zhang. 2022. PEPPi: Whole-proteome Protein-protein Interaction Prediction through Structure and Sequence Similarity, Functional Association, and Machine Learning. *Journal of Molecular Biology* 434, 10 (2022), 167530. doi:10.1016/j.jmb.2022.167530
- [9] Yung-Chun Chang, Ming-Siang Huang, Yi-Hsuan Huang, and Yi-Hsuan Lin. 2025. The influence of prompt engineering on large language models for protein-protein interaction identification in biomedical literature. 15, 1 (2025), 15493. doi:10.1038/s41598-025-99290-4
- [10] Ryan Engel and Gilchan Park. 2024. Evaluating Large Language Models for Predicting Protein Behavior under Radiation Exposure and Disease Conditions. In *Proceedings of the 23rd Workshop on Biomedical Natural Language Processing*. 427–439. doi:10.18653/v1/2024.bionlp-1.34
- [11] Noelia Ferruz et al. 2023. From sequence to function through structure: Deep learning for protein design. 21 (2023), 238–250. doi:10.1016/j.csbj.2022.11.014
- [12] Lixin Gong et al. 2024. LILRB4 represents a promising target for immunotherapy by dual targeting tumor cells and myeloid-derived suppressive cells in multiple myeloma. (2024). doi:10.3324/haematol.2024.285099
- [13] Graphistry, Inc. 2025. *Graphistry: GPU-accelerated Visual Graph Analytics*. <https://www.graphistry.com/>
- [14] Shobhit Jain and Gary D Bader. 2010. An improved method for scoring protein-protein interactions using semantic similarity within the gene ontology. 11, 1 (2010), 562. doi:10.1186/1471-2105-11-562
- [15] Minoru Kanehisa, Masahiro Furumichi, Yoko Sato, Yuki Matsuura, and Michihiro Ishiguro-Watanabe. 2025. KEGG: biological systems database as a model of the real world. *Nucleic Acids Research* 53 (2025), D672–D677. doi:10.1093/nar/gkae909
- [16] Olga Ksionda et al. 2012. Mechanism and function of Vav1 localization in TCR signaling. (2012), jcs.105148. doi:10.1242/jcs.105148
- [17] K. L Banta et al. 2022. Mechanistic convergence of the TIGIT and PD-1 inhibitory pathways necessitates co-blockade to optimize anti-tumor CD8+ T cell responses. *Immunity* 55 (2022), 512–526. doi:10.1016/j.immuni.2022.02.005
- [18] Dan Liu et al. 2025. PLM-interact: extending protein language models to predict protein-protein interactions. 9012 (10 2025). doi:10.1038/s41467-025-64512-w
- [19] N. Z. Malinga et al. 2022. Systemic levels of the soluble co-inhibitory immune checkpoints, CTLA-4, LAG-3, PD-1/PD-L1 and TIM-3 are markedly increased in basal cell carcinoma. *Translational Oncology* 19 (2022), 101384. doi:10.1016/j.tranon.2022.101384
- [20] Gilchan Park, Byung-Jun Yoon, Xihai Luo, Vanessa Lopez-Marrero, Shinjae Yoo, and Shantenu Jha. 2025. Comparative Performance Evaluation of Large Language Models for Extracting Molecular Interactions and Pathway Knowledge. *Journal of Computational Biology* 32, 7 (2025), 675–695. doi:10.1089/cmb.2025.0078
- [21] Madhusudan Paul and Ashish Anand. 2022. A New Family of Similarity Measures for Scoring Confidence of Protein Interactions Using Gene Ontology. *IEEE/ACM Transactions on Computational Biology and Bioinformatics* 19, 1 (2022), 19–30. doi:10.1109/TCBB.2021.3083150
- [22] V. Srinivasa Rao, K. Srinivas, G. N. Sujini, and G. N. Sunand Kumar. 2014. Protein-Protein Interaction Detection: Methods and Analysis. 2014, 1 (2014), 147648. doi:10.1155/2014/147648
- [23] Hasin Rehana, Nur Bengisu Çam, Mert Basmacı, Jie Zheng, Christianah Jemiyo, Yongqun He, Arzucan Özgür, and Junguk Hur. 2024. Evaluating GPT and BERT models for protein-protein interaction identification in biomedical text. 4, 1 (2024), vbae133. doi:10.1093/bioadv/vbae133
- [24] Trilochan Rout, Anjali Mohapatra, and Madhabananda Kar. 2024. A systematic review of graph-based explorations of PPI networks: methods, resources, and best practices. *Network Modeling Analysis in Health Informatics and Bioinformatics* 13, 29 (2024). doi:10.1007/s13721-024-00467-0
- [25] David Sehnal et al. 2021. Mol* Viewer: modern web app for 3D visualization and analysis of large biomolecular structures. *Nucleic Acids Research* 49, W1 (jul 2021), W431–W437. doi:10.1093/nar/gkab314
- [26] Paul Shannon et al. 2003. Cytoscape: A Software Environment for Integrated Models of Biomolecular Interaction Networks. *Genome Research* 13, 11 (2003), 2498–2504. doi:10.1101/gr.1239303
- [27] Rohit Singh, Kapil Devkota, Samuel Sledzieski, Bonnie Berger, and Lenore Cowen. 2022. Topsy-Turvy: integrating a global view into sequence-based PPI prediction. 38 (2022), i264–i272. Issue Supplement_1. doi:10.1093/bioinformatics/btac258
- [28] Samuel Sledzieski, Rohit Singh, Lenore Cowen, and Bonnie Berger. 2021. D-SCRIPT translates genome to phenotype with sequence-based, structure-aware, genome-scale predictions of protein-protein interactions. 12, 10 (2021), 969–982.e6. doi:10.1016/j.cels.2021.08.010
- [29] Damian Szklarczyk et al. 2021. The STRING database in 2021: customizable protein-protein networks, and functional characterization of user-uploaded gene/measurement sets. *Nucleic Acids Research* 49, D1 (2021), D605–D612. doi:10.1093/nar/gkaa1074
- [30] Susan Tweedie, Bryony Braschi, Kristian Gray, Tamsin E M Jones, Ruth L Seal, Bethan Yates, and Elspeth A Bruford. 2021. Genenames.org: the HGNC and VGNC resources in 2021. *Nucleic Acids Research* 49, D1 (2021), D939–D946. doi:10.1093/nar/gkaa980
- [31] Serbunt Unsul, Heval Atas, Muammer Albayrak, Kemal Turhan, Aybar Acar, and Tunca Dogan. 2022. Learning functional properties of proteins with language models. *Nature Machine Intelligence* 4 (03 2022), 227–245. doi:10.1038/s42256-022-00457-9
- [32] Christian von Mering, Roland Krause, Berend Snel, Michael Cornell, Stephen G. Oliver, Stefan Fields, and Peer Bork. 2002. Comparative assessment of large-scale data sets of protein-protein interactions. *Nature* 417, 6887 (2002), 399–403. doi:10.1038/nature750
- [33] Xiaoli Xue, Wei Zhang, and Anjing Fan. 2023. Comparative analysis of gene ontology-based semantic similarity measurements for the application of identifying essential proteins. 18, 4 (2023), e0284274. doi:10.1371/journal.pone.0284274
- [34] Tadashi Yokosuka et al. 2012. Programmed cell death 1 forms negative costimulatory microclusters that directly inhibit T cell receptor signaling by recruiting phosphatase SHP2. 209, 6 (2012), 1201–1217. doi:10.1084/jem.20112741

APPENDIX

A LLM Prompts for PPI Prediction

Initial candidate PPI predictions are generated using the following prompt templates:

System Prompt:

You are a helpful assistant specialized in molecular biology and protein interactions. When asked about protein interactions, provide clear, concise lists of interacting proteins with brief explanations. Focus on generating accurate protein names that are commonly used in databases and literature.

User Prompt:

List proteins that might interact with {protein}. Please provide a simple list of protein names (gene symbols/names) that could potentially interact with {protein}, along with a brief reason for each interaction (e.g., forms a complex, enzymatic substrate, signaling pathway, binding partner). Focus on well-known, documented interactions. Format your response as: PROTEIN_NAME - brief description of interaction type.

B Measures

B.1 Resnik SSM for Functional Similarity

Resnik semantic similarity measure (SSM) quantifies how much biological information is shared between two proteins (p_1, p_2). This measure computes the information content (IC) of the most informative common ancestor (MICA) between the GO annotation terms of each protein:

$$SSM_{\text{Resnik}}(p_1, p_2) = \max_{t \in C} IC(t) = IC(\text{MICA}(p_1, p_2)) \quad (1)$$

where C is the common ancestor terms of p_1 and p_2 . $IC(t) = -\ln(p(t))$ and $p(t)$ is the probability of term t occurring in a corpus of annotations. For each annotation term, we evaluate all annotated GO term pairs ($t_i^{(1)}, t_j^{(2)}$) associated with p_1 and p_2 , compute the corresponding $\text{MICA}(t_i^{(1)}, t_j^{(2)})$, and retain the term maximizing IC as the Resnik score for that annotation term. The information content of this most specific shared ancestor reflects the degree of functional relatedness between p_1 and p_2 . Higher IC indicates stronger plausibility of interaction, while lower IC values indicate greater uncertainty. A higher SSM indicates a higher likelihood that the predicted interaction is biologically meaningful.

B.2 D-SCRIPT for Physical Interaction Probability

We evaluate the probability of a physical interaction taking place between a predicted PPI pair using the deep learning method D-SCRIPT [28]. Unlike traditional methods that rely on homology, D-SCRIPT and its network-enhanced variants, [27], are capable of predicting novel, *zero-shot* interactions by learning the underlying biophysical constraints of protein docking. D-SCRIPT uses a pre-trained protein language model generates high-dimensional feature representations for each sequence. These embeddings are then processed by a structure-based protein interaction model to compute a binding compatibility score from the respective structures. In our implementation, we use the Topsy-Turvy model weights [27] an extension of the D-SCRIPT architecture that integrates global protein-protein interaction (PPI) network topology with local sequence information. Beyond the aggregate interaction probability

$P \in [0, 1]$, the module outputs a predicted inter-protein contact map $\hat{C}$. This depicts the probability of residue i from protein p_1 being in contact with residue j of protein p_2 , visualized as a 2D heatmap (Figure 4-5). This combination provides a structurally-informative, interpretable score to each prediction assessment, enabling users to further validate interactions with little PPI information.

C Implementation Details

Our platform is build on a decoupled, asynchronous architecture to ensure high-performance rendering and reproducibility. The system consists of three primary components:

- (1) **Visualization Web App:** An environment that manages the interactive 3D graph processor, LLM response processing, STRING and UniProt API calls, and Resnik similarity measures.
- (2) **LLM Inference Engine:** For generating interaction predictions at scale, we utilize an Ollama server hosting Llama 3.1 (8B parameters). This engine processes natural language prompts to predict potential protein-protein interactions and provide biological reasoning.
- (3) **Structural Inference Engine:** A dedicated Python-based Flask server hosting the Topsy-Turvy model weights. This module processes amino acid sequences to compute binding probabilities and residue-contact maps.

While D-SCRIPT can leverage GPU acceleration, our implementation demonstrates that the system remains highly responsive on standard consumer-grade CPU resources. By isolating the LLM inference and structural computations into dedicated microservices, we maintain a fluid user experience; the visualization remains interactive while complex structural and textual evidence is fetched asynchronously in the background.

D Preliminary Study

To evaluate the usability and functional requirements of our framework, we conducted a formative expert-led walkthrough with nine bioinformatics specialists representing diverse backgrounds (see Table 2). The experts were tasked with generating and identifying unverified but biologically plausible PPI candidates using our tool, and were asked to provide verbal feedback on the workflow and interface design. They subsequently compared this workflow to standard text-based LLM interactions.

This formative study led to several key improvements in the system architecture and visual interface. While the initial design lacked structural assessment, multiple experts requested physical docking probability, leading to the integration of the D-SCRIPT module. Experts also requested the ability to configure the GO parameters and base PPI network. Interface refinements included subnetwork pruning to manage graph density, filtering edges by interaction probability, and the integration of binding site metadata.

Experts (E1, E6) noted the system utility in exploring multiple PPI clusters uniformly, whereas text-based LLM interactions lack spatial context necessary for network exploration. Several experts (E2, E3, E4, E6, E9) observed that the combination of PPI analysis tools (3D network, multi-modal scoring, protein structure viewer) was more helpful in completing the task than the disjointed use of individual tools. Notably, multiple experts (E2, E3, E5, E6) highlighted

the importance of automatically distinguishing between grounded (database-supported) and unverified (absent from databases) interactions, which is obscured in raw outputs in text-based workflows.

D.1 Expert Backgrounds

Table 2 summarizes expert demographics for the preliminary walk-through of our system that informed our final design. Experts were recruited from the bioinformatics research community and represent a wide range of disciplinary backgrounds, professional experience, and familiarity with large language models and network visualization tools.

Table 2: Summary of expert demographics.

#	Degree, Major	Area	Yrs	LLMs	Vis
E1	PhD, Chem. Eng.	Protein MD	3	Exp.	Exp.
E2	MS, Comp. Sci.	ML, Bioinf.	9	Min.	Min.
E3	MS, Statistics	Bioinf., Soft. Eng.	8	Prof.	Exp.
E4	MS, Comp. Sci.	Bioinf., ML	5	Adv.	Min.
E5	MS, Comp. Sci. + Bioinf.	Bioinf., ML	25	Prof.	Prof.
E6	MS, Bioinf.	Genomics	6	Prof.	None
E7	PhD, Immunology	Small molecules	2	Min.	None
E8	PhD, Genome Stability	Molecular genetics	7	Min.	Min.
E9	PhD, Molecular Medicine	Radiation biology	10	Exp.	Min.

Note: Abbreviations for expertise levels: **Min.** = Minimal; **Prof.** = Proficient; **Exp.** = Experienced; **Adv.** = Advanced. **Yrs** = Years of domain experience. **Vis** = Familiarity with graph visualization tools. **MD** = molecular dynamics. **ML** = machine learning.

E Performance Evaluation

To evaluate the system’s responsiveness and scalability, we benchmarked the end-to-end latency of the “Analysis” stage across varying PPI candidate batch sizes ($N \in \{1, 5, 10, 20\}$). Using the same source protein (PDCD1) and LLM (Llama 3.1). These experiments were run on a 12-core Apple M2 Max CPU with 64GB of unified memory. Our pipeline demonstrated a significant reduction in overall latency by utilizing a concurrent architecture to execute analysis modules in parallel. With the exception of $N = 1$ (from concurrency overhead in thread pool initialization), all PPI batch sizes showed improvement with our enhanced pipeline as N increases (Table 3).

Table 3: End-to-end latency benchmarks of Analysis stage (STRING, D-SCRIPT, Resnik).

Batch (N)	Baseline (s)	Our Pipeline (s)	Improvement (%)
1	3.90	5.60*	—
5	13.87	11.61	16.3%
10	26.03	16.14	38.0%
20	53.18	38.77	27.1%

*Note: Nominal concurrency overhead at $N = 1$.

F Algorithms

Algorithm 1 presents the pseudocode for computing the similarity between the GO terms of two proteins. We compute similarity independently within the Molecular function (MF), Biological process (BP), and Cellular component (CC) annotation terms.

Algorithm 1 Resnik SSM Between Two Proteins

Require: Proteins A, B with annotated GO terms; GO DAG; Information Content (IC) for GO terms

Ensure: Resnik similarity scores for BP, MF, CC

```

1: for each annotation  $ns$  in {BP, MF, CC} do
2:    $A\_terms \leftarrow$  set of GO terms annotated to protein  $A$  in  $ns$ 
3:    $B\_terms \leftarrow$  set of GO terms annotated to protein  $B$  in  $ns$ 
4:   if  $A\_terms$  is empty or  $B\_terms$  is empty then
5:     similarity score for  $ns \leftarrow 0$ 
6:     continue to next annotation
7:   end if
8:    $scored\_pairs \leftarrow \emptyset$ 
9:   for each GO term  $a$  in  $A\_terms$  do
10:    for each GO term  $b$  in  $B\_terms$  do
11:       $MICA \leftarrow$  most informative common ancestor of  $a$  and  $b$ 
12:       $IC\_mica \leftarrow$  IC of  $MICA$ 
13:      add  $(a, b, MICA, IC\_mica)$  to  $scored\_pairs$ 
14:    end for
15:   end for
16:   if  $scored\_pairs$  is empty then
17:     similarity score for  $ns \leftarrow 0$ 
18:   else
19:     Sort  $scored\_pairs$  by  $IC\_mica$  descending
20:      $(go\_a, species\_a, go\_b, species\_b, mica, mica\_ic) \leftarrow$  first element of  $scored\_pairs$ 
21:      $mica\_ic\_norm \leftarrow$  normalized  $mica\_ic$ 
22:   end if
23: end for
24: return BP, MF, CC terms and scores

```